

KENNETH CAMACHO

Miami, FL • 786-376-7044 • kenneth.camacho.swe@gmail.com • linkedin.com/in/kennethcxv • github.com/kennethcxv • kennethcamacho.com

EDUCATION

Georgia Institute of Technology | M.S. Computer Science Atlanta, GA | Expected May 2027
Specialization: Machine Learning | Coursework: Distributed Systems, Advanced Operating Systems, Database Internals GPA: 3.9 / 4.0

Florida State University | Bachelor's Computer Science, *summa cum laude* Tallahassee, FL | Graduated: Aug 2026
Dean's List all semesters • Coursework: Operating Systems, Compilers, Algorithms, Networks, Architecture GPA: 3.8 / 4.0

WORK EXPERIENCE

NVIDIA | Systems Software Engineer Intern Santa Clara, CA | Sep 2026 – Present

- Built and optimized C++ systems components for NVIDIA DRIVE, cutting p95 execution latency **18%** by profiling CPU hotspots, reducing unnecessary memory copies, and tightening critical-path code.
- Developed Linux-based tooling and runtime support for performance-sensitive automotive workloads, reducing CPU utilization **12%** and improving throughput **21%** in internal benchmark scenarios.
- Added tracing, diagnostics, and automated stress tests for a multi-process software stack, shortening debugging time **~30%** and catching concurrency and reliability regressions before integration.

Rubrik | Software Engineer Intern Palo Alto, CA | May 2026 – Aug 2026

- Built and optimized React/TypeScript workflows for cloud-security alert triage and backup monitoring across multi-cluster environments, cutting p95 page-load latency **30%** through code-splitting, React Suspense, and targeted render-path optimization.
- Refactored GraphQL/Apollo data access across 4 production dashboards into shared hooks and normalized caching, eliminating **~50% of duplicated request logic** and fixing cross-view consistency issues affecting customer-facing security and backup workflows.

Chime | Software Engineer Intern San Francisco, CA | May 2025 – Dec 2025

- Shipped TypeScript/Node.js features in a payment microservice processing \$1.2M+ monthly transaction volume; cut p95 latency 20% via PostgreSQL query-plan tuning (EXPLAIN ANALYZE, partial indexes) and Kubernetes HPA scaling.
- Designed GraphQL/REST contracts and PostgreSQL schemas for a payments service consumed by 3 internal teams; authored the design doc, drove backward-compat review with risk and platform, and shipped Jest tests to 85% coverage.
- Cut launch-week payment incident MTTR **~40%** by adding structured logging, OpenTelemetry tracing spans, and Grafana SLO dashboards on the customer-facing checkout path.

PROJECTS

Distributed Key-Value Store (Raft) | Go, RocksDB, Docker Feb 2026 – Apr 2026

- Implemented Raft consensus in Go with leader election and log replication for a fault-tolerant key-value store across a 5-node cluster, validating strong consistency on 1M+ entries under simulated network partitions.
- Used RocksDB-backed LSM-tree storage with MVCC versioning; benchmarked **~10k writes/sec**, instrumented Prometheus metrics (ops/sec, p99 latency), and Dockerized each node.

Distributed Job Queue | Go, Redis, Docker, Linux Jan 2026 – Apr 2026

- Built a distributed job queue in Go with priority scheduling, token-bucket rate limiting, retries, and a dead-letter queue, scaling to 5 worker nodes processing 10k tasks/hour.
- Instrumented Prometheus metrics (queue depth, task latency, failure rate) and Grafana dashboards with alerting on stuck workers; documented runbooks for triaging throughput and reliability issues.

Semantic Code Search Engine | Python (FastAPI), PostgreSQL/pgvector, Redis, Docker Jan 2026 – Mar 2026

- Built a semantic code-search service over a 500k-line codebase, combining BM25 keyword retrieval with code embeddings in PostgreSQL/pgvector; reduced irrelevant top-k results 40% on a hand-labeled set vs. a vector-only baseline.
- Engineered an offline evaluation harness scoring recall@k and MRR on a labeled query set, plus a Redis-cached embedding layer cutting p95 query latency 35% under load.

AWARDS & LEADERSHIP

- Founder, Code Culture (Aug 2022 – Present)** | 4,500+ member student programming community; led 8 hackathons and 15+ workshops on algorithms, systems, web dev, and ML, mentoring 200+ students.
- 1st Place, EMerge Americas Hackathon (UM)** | Built a real-time fraud-detection prototype using Kafka, Python, and a gradient-boosted classifier; benchmarked anomaly detection on 100K simulated transactions.
- 1st Place, NextEra Energy Hackathon (FSU)** | 24-hour solar generation forecasting model (LSTM with weather and seasonality features) cutting MAPE 18% vs. a persistence baseline on NextEra's public dataset.
- 1st Place, Pluto Hacks** | In-browser Python/JavaScript playground via WebAssembly, with collaborative editing using Yjs CRDTs.

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, Python, Go, Java, SQL
Frontend: React, Next.js, Apollo, Tailwind CSS, HTML, CSS
Backend: Node.js, Express, FastAPI, gRPC, GraphQL, REST APIs, Microservices
Data & Search: PostgreSQL, pgvector, Redis, RocksDB, embeddings, semantic search, hybrid retrieval (BM25 + dense), vector databases
Cloud & DevOps: AWS, Docker, Kubernetes, Linux, CI/CD (GitHub Actions), Bazel, Prometheus, Grafana, Git, Kafka
Concepts: Distributed Systems, System Design, Concurrency, Operating Systems, Algorithms, Data Structures